Luiss
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degli Studi Sociali Guido Carli

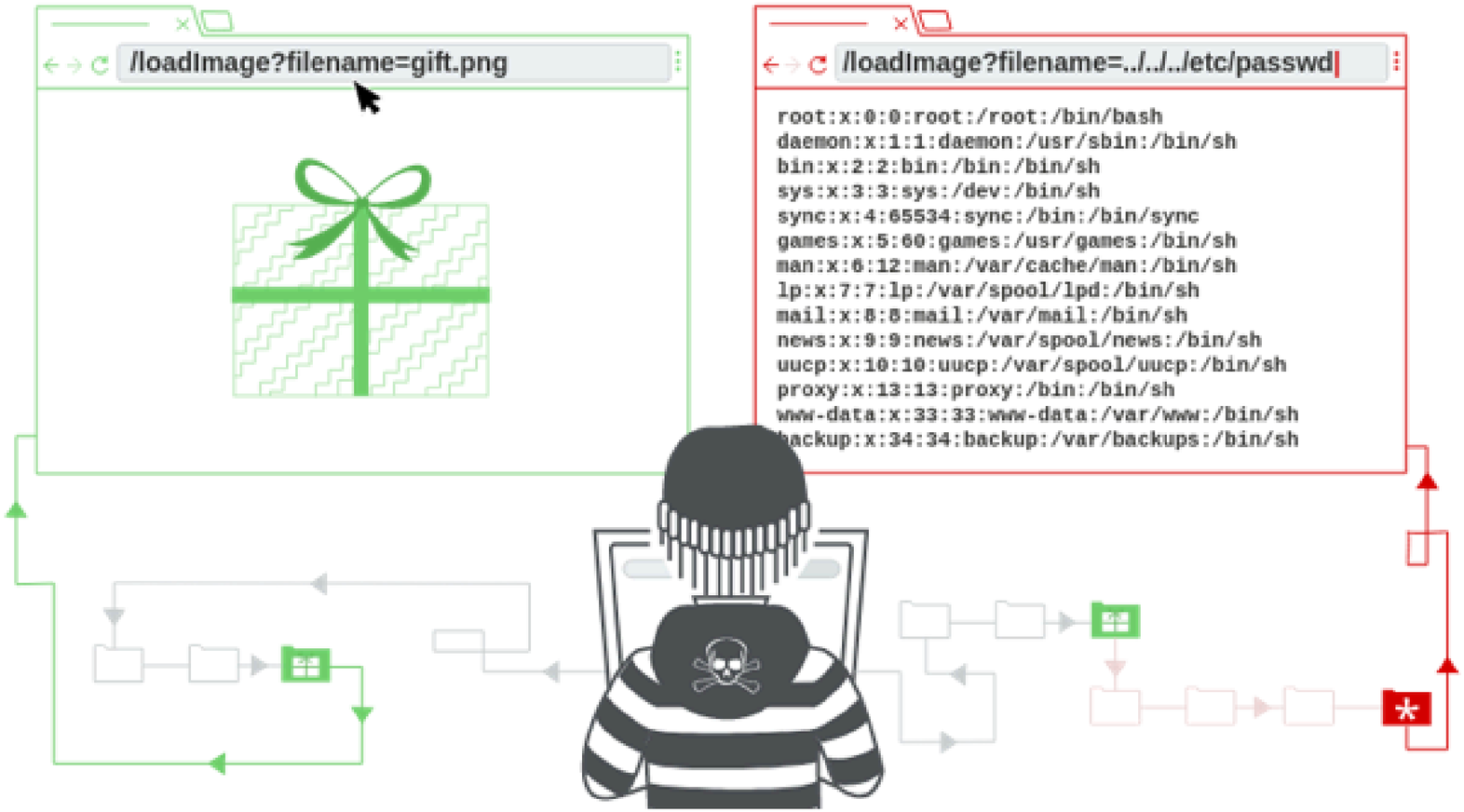
Path Traversal

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Path Traversal in a Nutshell



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- The webroot is the topmost directory in which the files of a website are stored
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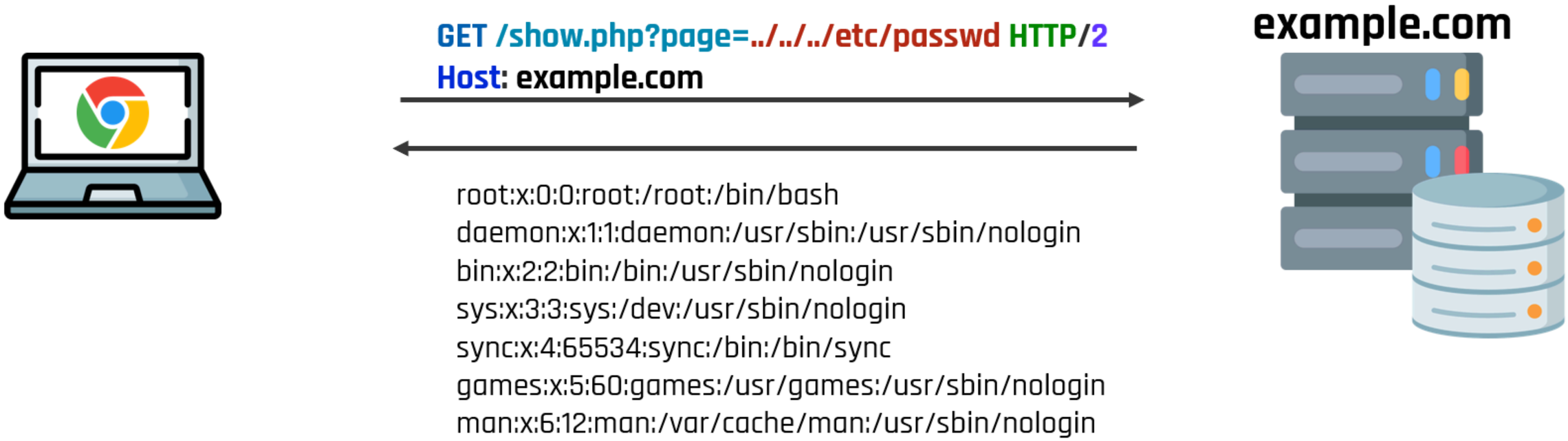
The webroot contains the file show.php above and a directory pages containing some text files that can be included by the PHP script

Intended Usage

Attack

Root cause of the problem: The user input provided through via page variable is not (correctly) filtered!

Attacker can “climb up” multiple levels in the directory hierarchy (and exit the webroot) by using ../ (Linux) or ..\ (Windows) and get access to any file on the web server (sensitive operating system files, TLS keys, etc.)



Preventing Path Traversals

- Ideally: don't use user controlled input as (part of) filenames
- In the real world: Validate all user inputs!
  - If possible, allow only a (static) list of file paths
  - Otherwise, compute the canonical path of the required file and ensure it is not outside the webroot (or the expected directory)

```php
<?php
    $pdir = "/var/www/html/pages/";
    $file = realpath($pdir . $_GET["file"]);

    if ($file !== false && strncmp($file, $pdir, strlen($pdir)) === 0) {
        echo file_get_contents($file);
    } else {
        echo "Error: invalid input";
    }
?>
```

Preventing Path Traversals - Defense in Depth

Reduced privileges of web server

- Restrict access of web server to its own directory
- Use sandbox environment (chroot jail, SELinux, containers,...) to enforce boundary between web server and the OS

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This is a so-called defense-in-depth mechanism: it is a good idea to deploy it, but it should not be the only adopted defense mechanism!

Important directories

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- **Server Root**: Contains logs & configurations. A few scripts put here their working directory

File permissions

Be aware of permissions given to directories. In particular:

- document root, containing HTML documents
- server root, containing log & configuration files; often CGI scripts run here
- the Common Gateway Interface (CGI) is a standard (RFC3875) that defines how Web server software can delegate the generation of Web pages to a console application. Such applications are known as CGI scripts; they can be written in any programming language, although scripting languages are often used

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It is a good idea to purposely define user and group for the web server

- e.g., www & `wwwgroup`
- HTML authors should be added to `wwwgroup`
- the www should have access only to the right files

Other configurations

Web servers may have additional capabilities, that can increase the risk

- automatic directory listing: an attacker can see the content of directories... what it we have left some sensitive data (e.g., our editor has left a temp copy of our PHP file?), symbolic links, development logs, source code control directories.
- symbolic link following: it may allow an attacker to access unexpected directories
- server side include (SSI): ".shtml" dynamic pages in the 90s... directives for including other files and executing commands. Still enabled somewhere.
- user maintained directory: Still used in several organization, e.g., `example.com/~user`

Fun With Flags

Web Challenge 01

Fun With Flags

```php
<?php
  highlight_file(__FILE__);
  $lang = $_SERVER['HTTP_ACCEPT_LANGUAGE'] ?? 'ot';
  $lang = explode(',', $lang)[0];
  $lang = str_replace('../', '', $lang);
  $c = file_get_contents("flags/$lang");
  if (!$c) $c = file_get_contents("flags/ot");
  echo '<img src="' . base64_encode($c) . '">';
```

Analysis

The application wants to show a flag based on the user's language

The user's language is sent by the browser with header HTTP_ACCEPT_LANGUAGE

The flag is retrieved from the flags directory. If missing, a global flag is used.

To prevent problems, `../` is replaced with ""

Analysis

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> What can go wrong?

Problems

- HTTP_ACCEPT_LANGUAGE is under the client control, hence it can be modified
- there is input sanitization on the value of this header but it is not very effective
- the value is used to access a path on the server
- hence, there is a user-controlled input that is used to build a file path
- the user can access any file that is accessible by the web server

HappySharedHostingWithSQLite

Web Challenge 04

HappySharedHostingWithSQLite

```php
<?php
  highlight_file(__FILE__);
  $ver = SQLite3::version();
  echo '<br />';
  echo $ver['versionString'] . '<br />';
  echo $ver['versionNumber'] . '<br />';

  $db = new SQLite3('file.db');
  $tablesquery = $db->query("SELECT name FROM sqlite_master WHERE type='table';");

  while ($table = $tablesquery->fetchArray(SQLITE3_ASSOC)) {
      echo $table['name'] . '<br />';
  }
?>
```

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Users

Analysis

The application show some data from a database.

The database is SQLite (all the information is contained in a single file).

There is no user input possible.

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There is no user input possible.

> What can go wrong?

> Where is the database stored?